Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 1 of 9 · Question 5 part a.ii · 3 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

<table>
<tr><td>5 (a) (i)</td><td>any four of

• exhale through tube A;
• time how long for limewater to go cloudy;
• replace limewater;
• inhale through tube B;
• compare time to go cloudy;</td><td></td><td>4 marks</td></tr>
<tr><td>(ii)</td><td>• more carbon dioxide in exhaled air;
• quicker/less time to go cloudy with exhaled air;
• produced by respiration;</td><td></td><td>3 marks</td></tr>
<tr><td>(b) (i)</td><td><table><tr><th>Lung volume</th><th>Letter</th><th>Volume in dm^3</th></tr><tr><td>tidal volume</td><td>W;</td><td>0.4;</td></tr><tr><td>vital capacity</td><td>Z;</td><td>2.25;</td></tr></table></td><td></td><td>4 marks</td></tr>
<tr><td>(ii)</td><td>• deep/maximum inhale;
• followed by maximum/forcefully exhale;</td><td>must be in correct order</td><td>2 marks</td></tr>
</table>

Total 13 marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 1 of 9 · Question 5 part a.ii · 3 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 2 of 9 · Question 1 part a · 3 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

1 (a)	• apparatus in correct order with T piece connected to both flasks; • long tubes in indicator; • short tubes above indicator;		3
(b)	limewater;	Allow sodium hydrogen carbonate	1
(c)	any four from: • breathe through mouth piece; • in and out; • 10 times/max of one minute; • compare contents of flasks; • exhaled flask should go cloudy if more carbon dioxide exhaled;		4
(d)	sterilise mouth piece/breath gently;		1
(e)	same volume of indicator in each flask;		1

Total 10 marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 2 of 9 · Question 1 part a · 3 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 3 of 9 · Question 1 part b · 1 mark

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

1 (a)	• apparatus in correct order with T piece connected to both flasks; • long tubes in indicator; • short tubes above indicator;		3
(b)	limewater;	Allow sodium hydrogen carbonate	1
(c)	any four from: • breathe through mouth piece; • in and out; • 10 times/max of one minute; • compare contents of flasks; • exhaled flask should go cloudy if more carbon dioxide exhaled;		4
(d)	sterilise mouth piece/breath gently;		1
(e)	same volume of indicator in each flask;		1

Total 10 marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 3 of 9 · Question 1 part b · 1 mark

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 4 of 9 · Question 2 part b.i · 1 mark

2023 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
2	(a)	(i)	nose/nasal cavity/mouth, trachea/windpipe, bronchi, bronchioles;;	one mark for two correct two marks for four correct	2
		(ii)	• large surface area; • thin epithelium/walls;	R cell wall/cell membrane	2
	(b)	(i)	B; 0.04% A, C and D are all incorrect figures		1
		(ii)	• 1200×16; • $\frac{19200}{1000}$ = 19 dm^3;	ECF if any value used from table Accept 19.2	2
		(iii)	• $\frac{20 \times 10\,500}{100}$; • $\frac{2100 \times 20}{100}$; • = 420 cm^3;	2 mark ECF	3
		(iv)	• more carbon dioxide in blood/lungs/body; • detected by (chemo)receptors (in arteries); • rate of breathing/tidal volume increases; • increases to remove carbon dioxide;		4

Total 14 marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 4 of 9 · Question 2 part b.i · 1 mark

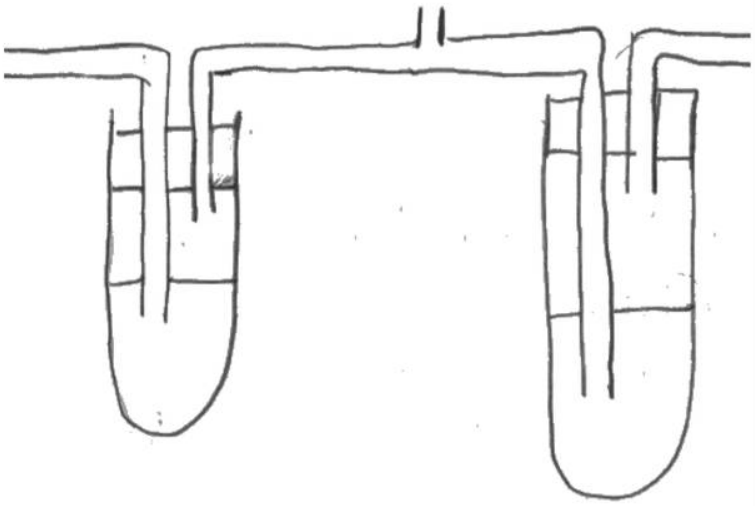
2023 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 5 of 9 · Question 4 part a.i · 3 marks

2024 JUN · Paper 01 · find part a.i in this mark scheme extract

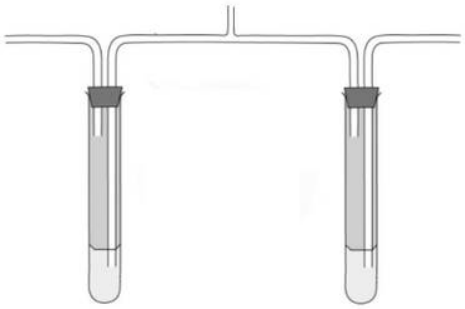
4	(a)	(i)	• apparatus correctly set up; • long tube in indicator; • short tube above indicator; 		(3)
		(ii)	Inhale and exhale gently/sterilise/clean mouthpiece/don't inhale liquid/care in assembling apparatus to avoid breaking glass;		(1)
	(b)	(i)	exercise/rest for the same length of time/at the same pace for each repeat/same student;		(1)
		(ii)	any four from • breathing rate increases (with vigorous exercise); • more oxygen inhaled/supplied; • for aerobic respiration; • more energy released/more ATP; • breakdown of lactic acid; • produced by anaerobic respiration;	ignore ref CO_2	(4)

Total for Question 4 = 9 marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 6 of 9 · Question 7 part a.iii · 2 marks

2024 JUN · Paper 02 (regional R) · find part a.iii in this mark scheme extract

7 (a)	(i)	glucose ⟶ lactic acid/lactate;	ignore any ref to CO_2 and H_2O on right side of equation	(1)
	(ii)	$C_6H_{12}O_6 + 6O_2; \longrightarrow 6H_2O + 6CO_2;$		(2)
	(iii)	general diagram; tubing correct way round; 	labels not needed long short, long short short long, short long	(2)
	(iv)	• inhale and exhale/breathe through the mouthpiece; • at rest; • time how long it takes the limewater to go cloudy/milky; • repeat following exercise;		(4)
(b)		any five from • energy released from (breakdown of) glucose; • during (aerobic) respiration; • (energy used) to add phosphate group; • to convert ADP to form ATP; • energy is released from ATP breakdown of ATP releases energy; • breaking of phosphate bond/ATP hydrolysed/ATP breaks down into ADP + P(i); • to provide energy for muscle contraction;		(5)

Total for Question 7 = 14 marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 7 of 9 · Question 5 part a.i · 4 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

<table>
<tr><td>5 (a) (i)</td><td>any four of

• exhale through tube A;
• time how long for limewater to go cloudy;
• replace limewater;
• inhale through tube B;
• compare time to go cloudy;</td><td></td><td>4 marks</td></tr>
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Total 13 marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 7 of 9 · Question 5 part a.i · 4 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 8 of 9 · Question 1 part c · 4 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

1 (a)	• apparatus in correct order with T piece connected to both flasks; • long tubes in indicator; • short tubes above indicator;		3
(b)	limewater;	Allow sodium hydrogen carbonate	1
(c)	any four from: • breathe through mouth piece; • in and out; • 10 times/max of one minute; • compare contents of flasks; • exhaled flask should go cloudy if more carbon dioxide exhaled;		4
(d)	sterilise mouth piece/breath gently;		1
(e)	same volume of indicator in each flask;		1

Total 10 marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 8 of 9 · Question 1 part c · 4 marks

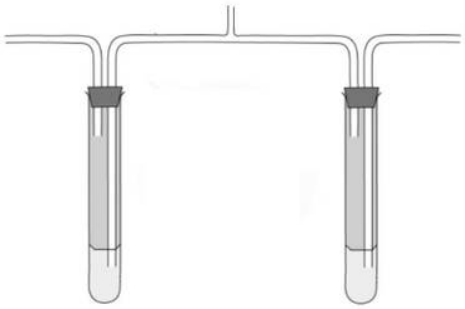
2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Set up and explain the indicator practical comparing carbon dioxide in breath

Answer 9 of 9 · Question 7 part a.iv · 4 marks

2024 JUN · Paper 02 (regional R) · find part a.iv in this mark scheme extract

7 (a)	(i)	glucose ⟶ lactic acid/lactate;	ignore any ref to CO_2 and H_2O on right side of equation	(1)
	(ii)	$C_6H_{12}O_6 + 6O_2; \longrightarrow 6H_2O + 6CO_2;$		(2)
	(iii)	general diagram; tubing correct way round; 	labels not needed long short, long short short long, short long	(2)
	(iv)	• inhale and exhale/breathe through the mouthpiece; • at rest; • time how long it takes the limewater to go cloudy/milky; • repeat following exercise;		(4)
(b)		any five from • energy released from (breakdown of) glucose; • during (aerobic) respiration; • (energy used) to add phosphate group; • to convert ADP to form ATP; • energy is released from ATP breakdown of ATP releases energy; • breaking of phosphate bond/ATP hydrolysed/ATP breaks down into ADP + P(i); • to provide energy for muscle contraction;		(5)

Total for Question 7 = 14 marks